

- b. What are the odds in favor of T ?
- c. What are the odds against T ?

7.8 The Addition Rule for Probability



Figure 7.35 Students can be sorted using a variety of possible categories like class year, major, whether they are a varsity athlete, and so forth. (credit: “Multicultural Mashup Melds Languages, Cultures at COD 36” by COD Newsroom/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify mutually exclusive events.
2. Apply the Addition Rule to compute probability.
3. Use the Inclusion/Exclusion Principle to compute probability.

Up to this point, we have looked at the probabilities of *simple* events. Simple events are those with a single, simple characterization. Sometimes, though, we want to investigate more complicated situations. For example, if we are choosing a college student at random, we might want to find the probability that the chosen student is a varsity athlete *or* in a Greek organization. This is a **compound event**: there are two possible criteria that might be met. We might instead try to identify the probability that the chosen student is *both* a varsity athlete *and* in a Greek organization. In this section and the [next](#), we'll cover probabilities of two types of compound events: those built using “or” and those built using “and.” We'll deal with the former first.

Mutual Exclusivity

Before we get to the key techniques of this section, we must first introduce some new terminology. Let's say you're drawing a card from a standard deck. We'll consider 3 events: H is the event “the card is a $\heartsuit$,” T is the event “the card is a 10,” and S is the event “the card is a $\spadesuit$.” If the card drawn is $J\spadesuit$, then H and T didn't occur, but S did. If the card drawn is instead $10\heartsuit$, then H didn't occur, but both T and S did.

We can see from these examples that, if we are interested in several possible events, more than one of them can occur simultaneously (both T and S , for example). But, if you think about all the possible outcomes, you can see that H and S can *never* occur simultaneously; there are no cards in the deck that are both $\heartsuit$ and $\spadesuit$. Pairs of events that cannot both occur simultaneously are called **mutually exclusive**. Let's go through an example to help us better understand this concept.

EXAMPLE 7.28

Identifying Mutually Exclusive Events

Decide whether the following events are mutually exclusive. If they are not mutually exclusive, identify an outcome that would result in both events occurring.

1. You are about to roll a standard 6-sided die. E is the event “the die shows an even number” and F is the event “the die shows an odd number.”
2. You are about to roll a standard 6-sided die. E is the event “the die shows an even number” and S is the event “the die shows a number less than 4.”
3. You are about to flip a coin 4 times. J is the event “at least 2 heads are flipped” and K is the event “fewer than 3 tails are flipped.”

 Solution

1. Let's look at the outcomes for each event: $E = \{2, 4, 6\}$ and $F = \{1, 3, 5\}$. There are no outcomes in common, so E and F are mutually exclusive.
2. Again, consider the outcomes in each event: $E = \{2, 4, 6\}$ and $S = \{1, 2, 3\}$. Since the outcome 2 belongs to both events, these are *not* mutually exclusive.
3. Suppose the results of the 4 flips are HTTH. Then at least 2 heads are flipped, and fewer than 3 tails are flipped. That means that both J and K occurred, and so these events are *not* mutually exclusive.

 YOUR TURN 7.28

Suppose you're about to draw one card from a deck containing only these 10 cards: $A\heartsuit, A\spadesuit, A\clubsuit, A\diamondsuit, K\spadesuit, K\clubsuit, Q\heartsuit, Q\spadesuit, J\heartsuit, J\spadesuit$. Decide whether these events are mutually exclusive:

1. E is the event "the card is an ace" and F is the event "the card is a king."
2. R is the event "the card is a $\heartsuit$ " and E is the event "the card is an ace."
3. R is the event "the card is a $\heartsuit$ " and F is the event "the card is a king."

The Addition Rule for Mutually Exclusive Events

If two events are mutually exclusive, then we can use addition to find the probability that one or the other event occurs.

FORMULA

If E and F are mutually exclusive events, then

$$P(E \text{ or } F) = P(E) + P(F).$$

Why does this formula work? Let's consider a basic example. Suppose we're about to draw a *Scrabble* tile from a bag containing A, A, B, E, E, E, R, S, S, U. What is the probability of drawing an E or an S? Since 3 of the tiles are marked with E and 2 are marked with S, there are 5 tiles that satisfy the criteria. There are ten tiles in the bag, so the probability is $\frac{5}{10} = \frac{1}{2}$. Notice that the probability of drawing an E is $\frac{3}{10}$ and the probability of drawing an S is $\frac{2}{10}$; adding those together, we get $\frac{3}{10} + \frac{2}{10} = \frac{5}{10}$. Look at the numerators in the fractions involved in the sum: the 3 represents the number of E tiles and the 2 is the number of S tiles. This is why the Addition Rule works: The total number of outcomes in one event or the other is the sum of the numbers of outcomes in each of the individual events.

EXAMPLE 7.29

Using the Addition Rule

For each of the given pairs of events, decide if the Addition Rule applies. If it does, use the Addition Rule to find the probability that one or the other occurs.

1. You are rolling a standard 6-sided die. Event A is "roll an even number" and event B is "roll a 3."
2. You are drawing a card at random from a standard 52-card deck. Event R is "draw a $\heartsuit$ " and event S is "draw a king."
3. You are rolling a pair of standard 6-sided dice. Event E is "roll an odd sum" and event F is "roll a sum of 10." The table we constructed in [Example 7.18](#) might help.

		First Die					
		1	2	3	4	5	6
Second Die	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Figure 7.36

✓ Solution

1. Since 3 is not an even number, these events are mutually exclusive. So, we can use the Addition Rule: since $P(A) = \frac{3}{6}$ and $P(B) = \frac{1}{6}$, we get $P(A \text{ or } B) = \frac{3}{6} + \frac{1}{6} = \frac{2}{3}$.
2. If the card drawn is $K\heartsuit$, then both R and S occur. So, they aren't mutually exclusive, and the Addition Rule doesn't apply.
3. Since 10 is not odd, these events are mutually exclusive. Since $P(E) = \frac{18}{36}$ and $P(F) = \frac{3}{36}$, the Addition Rule gives us $P(E \text{ or } F) = \frac{18}{36} + \frac{3}{36} = \frac{7}{12}$.

> YOUR TURN 7.29

Suppose you're about to draw one card from a deck containing only these 10 cards:

$A\heartsuit, A\spadesuit, A\clubsuit, A\diamondsuit, K\spadesuit, K\clubsuit, Q\heartsuit, Q\spadesuit, J\heartsuit, J\spadesuit$. If appropriate, use the Addition Rule to find the probability that one or the other of these events occurs:

1. E is the event "the card is an ace" and F is the event "the card is a king."
2. R is the event "the card is a $\heartsuit$ " and E is the event "the card is an ace."
3. R is the event "the card is a $\heartsuit$ " and F is the event "the card is a king."

Finding Probabilities When Events Aren't Mutually Exclusive

Let's return to the example we used to explore the Addition Rule: We're about to draw a *Scrabble* tile from a bag containing A, A, B, E, E, E, R, S, S, U. Consider these events: J is "draw a vowel" and K is "draw a letter that comes after L in the alphabet." Since there are 6 vowels, $P(J) = \frac{6}{10}$. There are 4 tiles with letters that come after L alphabetically, so $P(K) = \frac{4}{10}$. What is $P(J \text{ or } K)$? If we blindly apply the Addition Rule, we get $\frac{6}{10} + \frac{4}{10} = 1$, which would mean that the compound event $J \text{ or } K$ is certain. However, it's possible to draw a B, in which case neither J nor K happens. Where's the error?

The events are not mutually exclusive: the outcome U belongs to both events, and so the Addition Rule doesn't apply. However, there's a way to extend the Addition Rule to allow us to find this probability anyway; it's called the **Inclusion/Exclusion Principle**. In this example, if we just add the two probabilities together, the outcome U is included in the sum twice: It's one of the 6 outcomes represented in the numerator of $\frac{6}{10}$, and it's one of the 4 outcomes represented in the numerator of $\frac{4}{10}$. So, that particular outcome has been "double counted." Since it has been *included* twice, we can get a true accounting by *excluding* it once: $\frac{6}{10} + \frac{4}{10} - \frac{1}{10} = \frac{9}{10}$. We can generalize this idea to a formula that we can apply to find the probability of any compound event built using "or."

FORMULA

Inclusion/Exclusion Principle: If E and F are events that contain outcomes of a single experiment, then

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F).$$

It's worth noting that this formula is truly an extension of the Addition Rule. Remember that the Addition Rule requires that the events E and F are mutually exclusive. In that case, the compound event (E and F) is impossible, and so $P(E \text{ and } F) = 0$. So, in cases where the events in question are mutually exclusive, the Inclusion/Exclusion Principle reduces to the Addition Rule.

EXAMPLE 7.30**Using the Inclusion/Exclusion Principle**

Suppose we have events E , F , and G , associated with these probabilities:

$$P(E) = 0.45$$

$$P(F) = 0.6$$

$$P(G) = 0.55$$

$$P(E \text{ and } F) = 0.2$$

$$P(E \text{ and } G) = 0.2$$

$$P(F \text{ and } G) = 0.25$$

Compute the following:

1. $P(E \text{ or } F)$
2. $P(E \text{ or } G)$
3. $P(F \text{ or } G)$

✓ Solution

1. Using the Inclusion/Exclusion Principle, we get:

$$\begin{aligned} P(E \text{ or } F) &= P(E) + P(F) - P(E \text{ and } F) \\ &= 0.45 + 0.6 - 0.2 \\ &= 0.85. \end{aligned}$$

2. Again, we'll apply the Inclusion/Exclusion Principle:

$$\begin{aligned} P(E \text{ or } G) &= P(E) + P(G) - P(E \text{ and } G) \\ &= 0.45 + 0.55 - 0.2 \\ &= 0.8. \end{aligned}$$

3. Applying the Inclusion/Exclusion Principle one more time:

$$\begin{aligned} P(F \text{ or } G) &= P(F) + P(G) - P(F \text{ and } G) \\ &= 0.6 + 0.55 - 0.25 \\ &= 0.9. \end{aligned}$$

**YOUR TURN 7.30**

You are about to roll a special 6-sided die that has both a colored letter and a colored number on each face. The faces are labeled with: a red 1 and a blue A, a red 1 and a green A, an orange 1 and a green B, an orange 2 and a red C, a purple 3 and a brown D, an orange 4 and a blue E. Find the probabilities of these events:

1. The number is orange or even.
2. The letter is green or an A.
3. The number is even or the letter is green.

Check Your Understanding

You are about to draw a card at random from a deck containing only these 10 cards: $A\heartsuit, A\spadesuit, A\clubsuit, A\diamondsuit, K\spadesuit, K\clubsuit, Q\heartsuit, Q\spadesuit, J\heartsuit, J\spadesuit$. Compute the following probabilities:

44. You draw an ace or a king.
45. You draw a $\spadesuit$ or a $\clubsuit$.
46. You draw an ace or a $\heartsuit$.
47. You draw a jack or a $\heartsuit$.
48. You draw a jack or a $\clubsuit$.
49. You draw a king or a $\diamondsuit$.



SECTION 7.8 EXERCISES

For the following exercises, we are considering a special 6-sided die, with faces that are labeled with a number and a letter: 1A, 1B, 2A, 2C, 4A, and 4E. You are about to roll this die once.

1. What is the probability of rolling a 1 or a 2?
2. What is the probability of rolling a 4 or a B?
3. What is the probability of rolling an even number or a consonant?
4. What is the probability of rolling a 2 or an E?
5. What is the probability of rolling an odd number or a vowel?
6. What is the probability of rolling an odd number or a consonant?

In the following exercises, you are drawing a single card from a standard 52-card deck.

7. What is the probability that you draw a $\heartsuit$ or a $\spadesuit$?
8. What is the probability that you draw a $\heartsuit$ or a 5?
9. What is the probability that you draw a 2 or a 3?
10. What is the probability that you draw a card with an even number on it?
11. What is the probability that you draw a card with an even number on it or a $\clubsuit$?
12. What is the probability that you draw an ace or a king?
13. What is the probability that you draw a face card (king, queen, or jack)?
14. What is the probability that you draw a face card or a $\clubsuit$?

For the following exercises, use the table provided here, which breaks down the enrollment at a certain liberal arts college by class year and area of study:

		Class Year				
		First-Year	Sophomore	Junior	Senior	Totals
Area Of Study	Arts	138	121	148	132	539
	Humanities	258	301	275	283	1117
	Social Science	142	151	130	132	555
	Natural Science/Mathematics	175	197	203	188	763
	Totals	713	770	756	735	2974

15. What is the probability that a randomly selected student is a first-year or sophomore?
16. What is the probability that a randomly selected student is a junior or an arts major?
17. What is the probability that a randomly selected student is majoring in the social sciences or the natural sciences/mathematics?
18. What is the probability that a randomly selected student is a social science major or a sophomore?
19. What is the probability that a randomly selected student is a senior or is a humanities major?
20. What is the probability that a randomly selected student is majoring in the arts or humanities?

The following exercises are about the casino game roulette. In this game, the dealer spins a marble around a wheel that contains 38 pockets that the marble can fall into. Each pocket has a number (each whole number from 0 to 36,

along with a double zero) and a color (0 and 00 are both green; the other 36 numbers are evenly divided between black and red). Players make bets on which number (or groups of numbers) they think the marble will land on. The figure shows the layout of the numbers and colors, as well as some of the bets that can be made.

		0	00	
1 to 18 EVEN	1st Dozen	1	2	3
		4	5	6
		7	8	9
		10	11	12
	2nd Dozen	13	14	15
		16	17	18
		19	20	21
		22	23	24
	3rd Dozen	25	26	27
		28	29	30
		31	32	33
		34	35	36
19 to 36 ODD				
		2 to 1	2 to 1	2 to 1

Roulette Table (credit: "American Roulette Table Layout" by Film8ker/Wikimedia Commons, Public Domain)

What is the probability of winning at least one of the following pairs of bets on a single spin of the wheel?

21. First dozen (wins if any of the numbers 1–12 come up) or second dozen (wins on 13–24)
22. Red (wins on any of the 18 red numbers) or black (wins on any of the 18 black numbers)
23. Even (wins on any even number 2–36; 0 and 00 both lose this bet) or red
24. Middle column (the numbers 2, 5, 8, 11, ..., 35) or black
25. Middle column or red
26. Right column (the numbers 3, 6, 9, ..., 36) or black
27. Right column or red
28. Odd or black
29. Even or black
30. The street bet (a bet on 3 numbers that make up a row on the table) on 1, 2, 3 or odd
31. The street bet on 1, 2, 3 or even
32. The corner bet (a bet on 4 numbers that form a square on the table) on 1, 2, 4, 5 or first dozen
33. The corner bet on 1, 2, 4, 5 or second dozen
34. The basket bet (which wins on 0, 00, 1, 2, 3) or red
35. The basket bet or black